Divye Kalra

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Education

Johns Hopkins University — **Baltimore, USA**
M.S. Cybersecurity — *Aug 2024 – Dec 2025*

Birla Institute of Technology and Science (BITS) Pilani — **Hyderabad, India**
B.E. Electrical and Electronics; M.Sc. Mathematics (Double Major) — *Aug 2019 – Jun 2024*

Technical Skills

- **Languages & Core Technologies:** Python, C, C++, Java, Rust, SQL, Bash, HTML/CSS, AWS, Azure, Azure DevOps, GCP, Docker, Kubernetes (k8s), GKE, OpenStack, Linux, FastAPI, Django, REST API Design, Burp Suite, Wireshark, Ghidra, Git, GitHub, MySQL, PostgreSQL, Neo4j, ElasticSearch/OpenSearch, MinIO, Prometheus, Grafana, Loki, Tempo, Proxmox, Jira, Ollama, Terraform, GitOps, Flux, MCP, OpenTelemetry, LiteLLM, Airflow
- **Security:** Application Security, Secure SDLC, Threat Modeling, SIEM, Incident Response, Logging and Monitoring, Threat Intelligence, IAM/RBAC/MFA/SSO, Endpoint Security, Zero Trust Architecture, Risk Management and Compliance, MITRE ATT&CK, ISO 27001, CIS, HIPAA, PCI-DSS

Experience

GlobalLogic — Software Engineer Intern — June 2026 – Present

- Owned the multi-cloud infrastructure (GCP, AWS, Azure) end-to-end behind a company-wide knowledge engine serving internal teams and meeting the needs of external clients – a platform for semantic, contextual, graph, and agentic search handled using OpenSearch over data in Postgres, MinIO, and Neo4j, with Airflow orchestrating ingestion.
- Replaced an 11-step manual deployment with GitOps-driven CI/CD and infrastructure-as-code (Terraform + Flux), accelerating product delivery and giving every backend infrastructure change a reviewed, auditable path to production; automated TLS issuance via ZeroSSL and DNS configuration, and instrumented the full stack with Prometheus, Grafana, Loki, and Tempo.
- Hardened the platform end-to-end – least-privilege IAM and Kubernetes RBAC, managed secrets and customer-managed encryption keys in all three clouds (Key Vault, Secrets Manager, Cloud KMS), and image/dependency scanning gated in CI – then troubleshot and drove flagged CVEs to closure service by service.

Wildlife for All — Information Security Engineer — Feb 2026 – June 2026

- Audited Google Workspace and Slack against CIS benchmarks, documenting 40+ misconfigurations spanning identity, email domain authentication, third-party app grants, external sharing, and offboarding – attention to detail mapping each to a control, severity rating, and concrete fix
- Remediated the highest-severity items first: enforced admin MFA, removed unused OAuth connectors from 65 Slack integrations, restricted Drive public sharing, and progressed Gmail DMARC toward enforcement – closing the top account-takeover and data-exfiltration paths.

JHU Applied Physics Laboratory — Information Security Engineer — Jan 2025 – Dec 2025

- Designed and implemented an SCMS (Secure Credential Management System) reference in Python — a 6-service system with a 3-tier CA hierarchy, signed V2X message broadcast, and a roadside server running 15+ real-time misbehavior checks against the SAE J3161 standard.
- Mitigated replay and location-spoofing attacks in V2X networks via a pseudonym certificate revocation pipeline using cryptographic linkage values; integrated C-V2X into an open-source platform, cutting CRL delivery latency by 20% vs. baseline DSRC broadcast.

IITB Trust Lab — Applied Cryptography Engineer — Jan 2024 – Jun 2024

- Delivered the first open-source Rust implementation of CASE, cutting encryption overhead ∼30% via a hybrid symmetric/asymmetric fixed-size key design and accelerating anonymous signature verification ∼15% by batching hash computations and eliminating redundant scalar multiplications.

Projects

- **Production-Style Homelab Infrastructure**
  Fully self-hosted home infrastructure — deploying **10+** services on Proxmox VMs and containers including TrueNAS, Pi-hole, OPNsense, and a metrics-and-uptime alerting stack — with Tailscale VPN for remote LAN access without port-forwarding and select workloads on AWS, eliminating all cloud storage subscriptions.
- **LLM-Powered Security Code Review Assistant**
  Built and shipped a two-phase AI agent pipeline — regex pattern matching to surface candidates, GPT to reason about exploitability in context and generate secure fixes — detecting 25+ vulnerability classes in 8+ languages at 95% accuracy; exposed as a customer-facing FastAPI web UI and CLI with every result mapped to OWASP, CWE, MITRE ATT&CK, and CVSS.
- **SBOM-Driven Runtime Admission Controller**
  Engineered a software supply-chain pipeline: build-time SBOM generation, an admission webhook that intercepts pod creation and cross-references image manifests against EPSS exploitability scores, and dynamic OPA/Rego policy generation — threat-modeled against XZ Utils-style backdoors, with Sigstore/Cosign signing and verification.
- **Kubernetes Security (Policies, Runtime, and Exploitation)**
  Executed full Kubernetes attack chains to problem solve real exploitation paths – Server-Side Request Forgery (SSRF) to steal a mounted service account token and enumerate cluster secrets, over-permissioned role bindings to exfiltrate credentials via pod volume mount, and a privileged container escape onto the underlying nodes — then closed each path with Kyverno (policy-as-code) admission rules aligned to the CIS Kubernetes Benchmark, validating detection coverage with Cilium Tetragon kernel-level eBPF telemetry.

Certifications and Achievements

- AWS Certified Cloud Practitioner (AWS CCP)
- CompTIA Security+